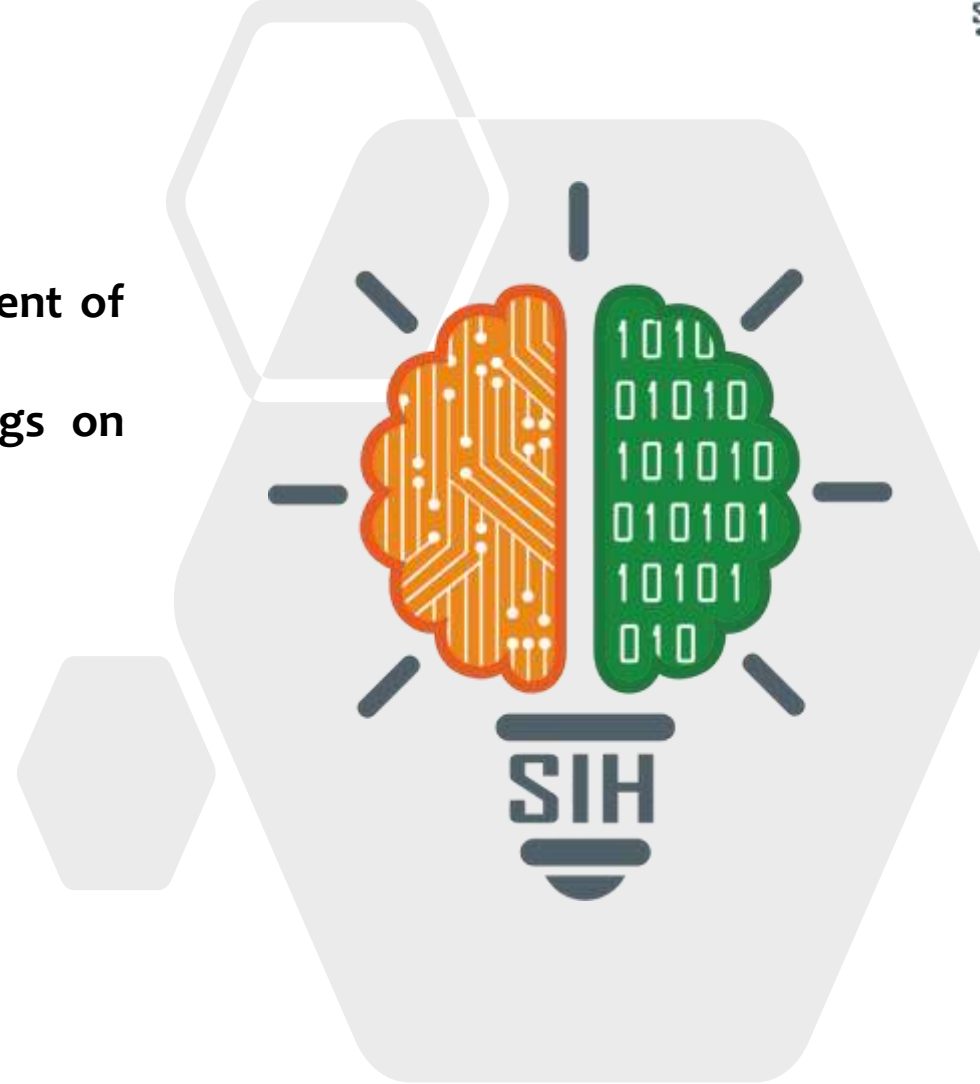


SMART INDIA HACKATHON 2025



- [Problem Statement ID](#) -25021
- [Problem Statement Title](#)- AI based development of Laser based QR Code marking on 'track fittings on Indian Railways'.
- [Theme](#)- Transportation & Logistics
- [PS Category](#)- Hardware
- [Team ID](#)-56924
- [Team Name](#)-TrackTaggers



AI-based development of Laser-QR Code marking on 'Track fittings for Indian Railways'

-Engraving code decoding safety

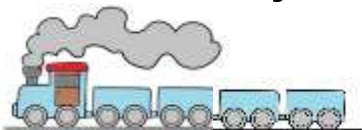


PROBLEM STATEMENT:

🚂 Indian Railways uses crores of fittings with **no unique identity**, leaving vendor details, warranty, inspection, and their performance untraceable. ⚠️ It leads to **errors, counterfeits, poor inventory, and safety risks** causing delays or accidents.

SOLUTION:

•**AI-guided lasers** etch tamper-proof **QR codes** on rail fittings (clips, pads, liners, sleepers) for real-time identification with **UDM integration** and it ensures precision, durability, and reliability even in harsh railway conditions

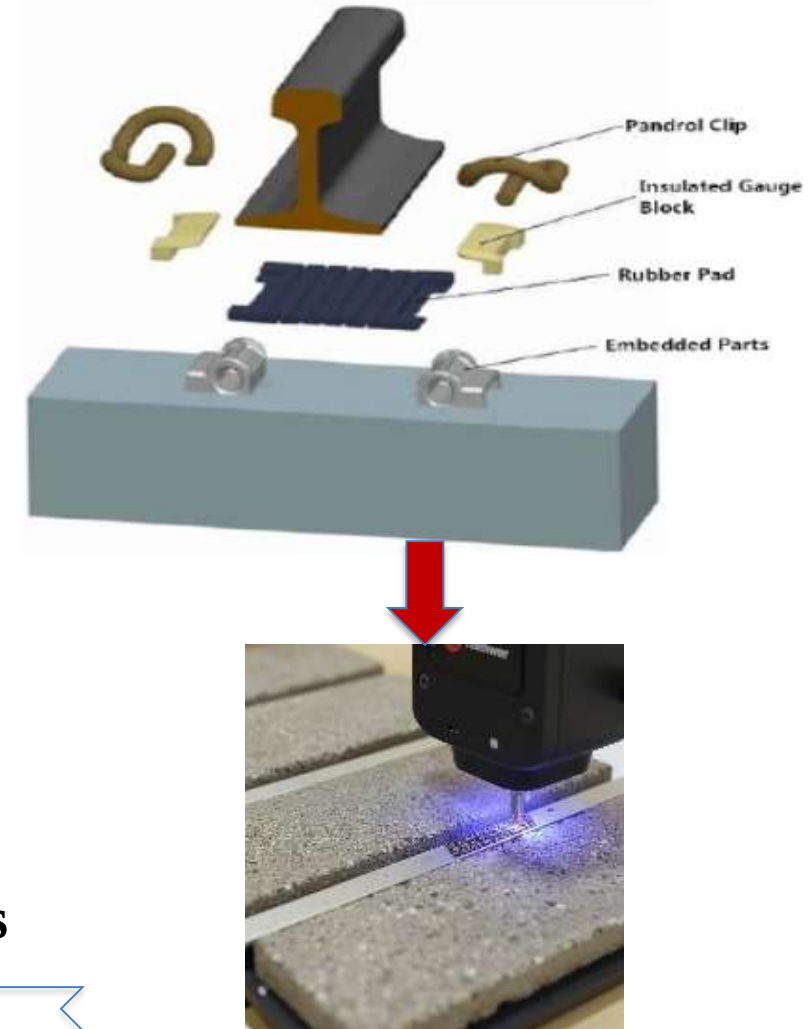


Target Audience

- Inspector
- Keyman
- Trackman
- Engineer
- Senior officers
- RDSO/ Workshops

[Solution explanation video](#)

Refer this link



TECHNICAL APPROACH

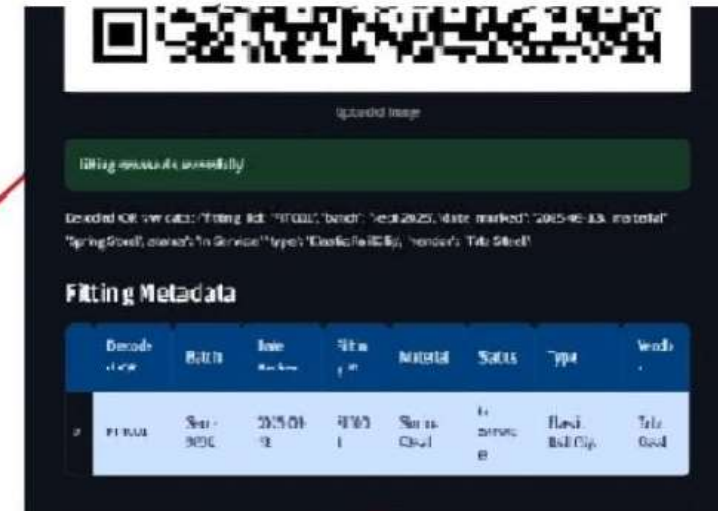
- Laser marking machine - Fiber ,CO₂ ,UV lasers
- AI- Open CV ,ML Kit (for low network areas) QR detection
- Mobile App - Flutter ,Android
- Backend -Flask ,Node.js with Database Firebase/ MySQL
- UDM Portal integration via secure APIs

Secures data with
Firebase Rules and
admin credential in
backend.

Supports real-time sync
Stores collections like
fittings and
quality issues



Scan and decode QR codes
to extract
metadata(fitting ID,batch,
material)

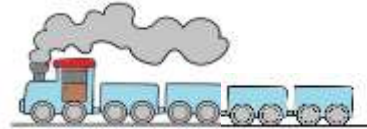


Persistent centralized
storage
accessible from anywhere

FEASIBILITY AND VIABILITY



Durable laser QR codes enable tamper-proof tracking, RDSO/UDM integration, quick inspections, nationwide scalability, and huge cost savings.



[Prototype video\(app\)](#) [Prototype\(web\)](#)

Refer these
links

Potential challenges

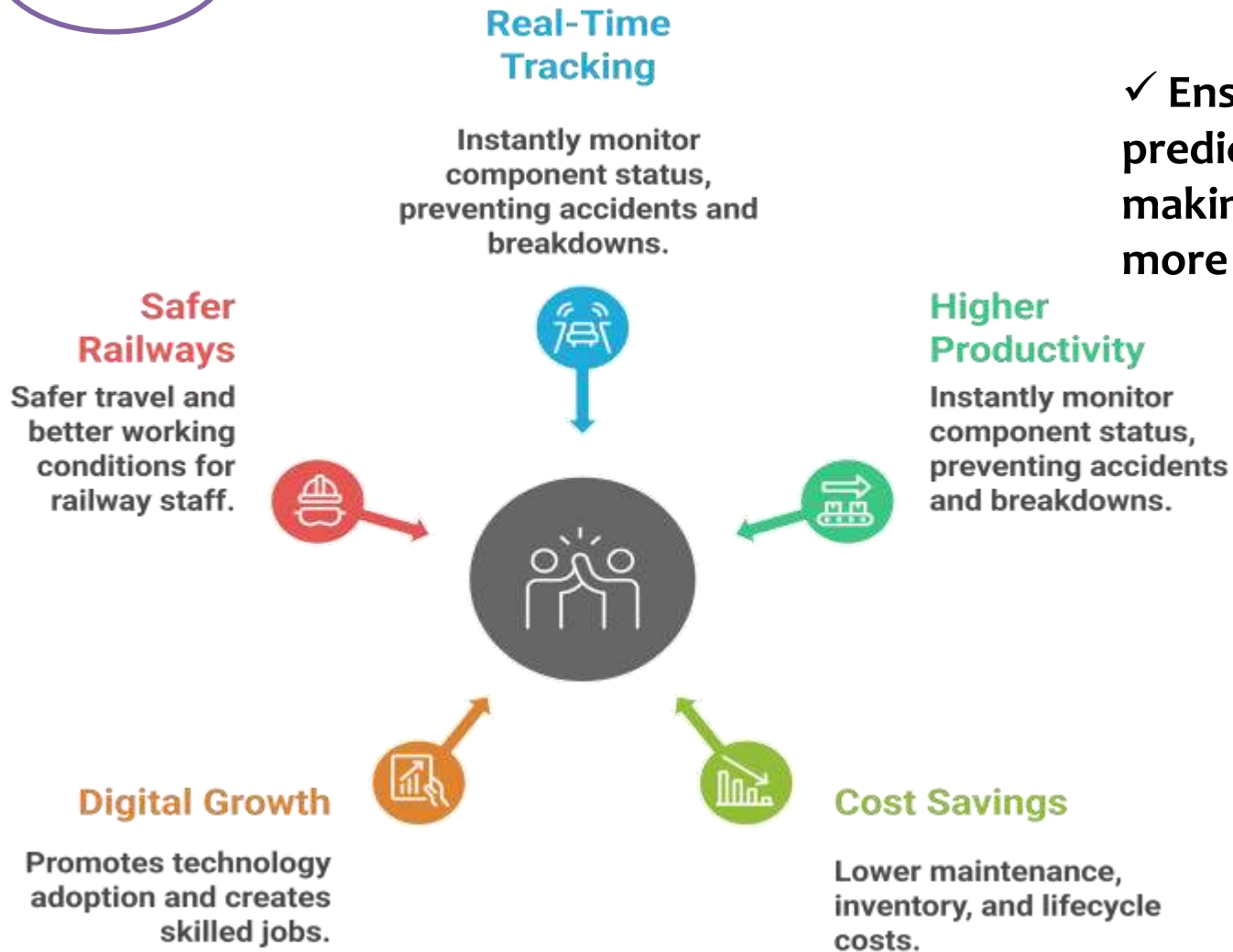
What if QR codes get damaged?
How to speed up inspections accurately?
Will high costs slow adoption?
What if vendors resist?
How to ensure long-term safety and efficiency?

Strategies to Overcome

- ✓ Durable QR for harsh conditions
- ✓ AI-assisted fast, accurate inspections
- ✓ Phased rollout to justify costs
- ✓ Vendor incentives for adoption
- ✓ Predictive analytics + planned maintenance

IMPACT AND BENEFITS

TRACK TAGGERS



✓ Ensures vendor accountability, faster audits predictive maintenance, and transparent procurement making the railway supply chain smarter, safer, and more cost-efficient.

Estimation From Statistics

METRICS	CAPITAL PER ANNUM(₹)
INITIAL INVESTMENT	13,19,87,025
SOFTWARE & DATA STORAGE	65,99,351.25
MAINTAINENCE & LABOUR	1,75,98,270
SAVINGS	10,55,89,620
RETURN ON INVESTMENT (ROI)	~1.5 years

RESEARCH AND REFERENCES



1. [All About Laser Etching in Metal Manufacturing](#) — CSMFG
2. [Mobile based Inventory Management System with QR code](#) –IEEE
3. [Laser-Based Railway Track Inspection-](#) AdorTech
4. [QR Code Condition / Surface Quality Detection](#) — Nyckel
5. [Recognition of Perspective Distorted QR Codes with a Partially Damaged Finder Pattern in Real Scene Images](#) — MDPI